

6

Multi-Pump Station Pipelines

In [Chapter 5](#) we calculated the total pressure required to pump a liquid through a pipeline from point A to point B at a specified flow rate. Three components of the total pressure required (friction head, elevation head, and delivery pressure) were analyzed. Depending on the maximum allowable operating pressure (MAOP) of the pipeline we concluded that one or more pump stations may be required to handle the throughput.

In this chapter we will discuss pipeline hydraulics for multiple pump stations. Hydraulic balance and how to determine the intermediate booster pump stations will be explained. To utilize pipe material efficiently we will explore telescoping pipe wall thickness and grade tapering. Also covered will be slack line flow and a more detailed analysis of batched pipeline hydraulics.

6.1 Hydraulic Balance and Pump Stations Required

Suppose calculations indicate that at a flow rate of 5000 gal/min, a 100 mile pipeline requires a pressure of 2000 psi at the beginning of the pipeline. This 2000 psi pressure may be provided in two steps of 1000 psi each or three steps of approximately 670 psi each. In fact, due to the internal pressure limit of the pipe, we may not be able to provide one pump station at the beginning of the pipeline, operating at 2000 psi. Most pipelines have an internal

pressure limit of 1000 to 1440 psi based on pipe wall thickness, grade of steel, etc., as we found in [Chapter 4](#). Therefore, in long pipelines the total pressure required to pump the liquid is provided in two or more stages by installing intermediate booster pumps along the pipeline.

In the example case with a 2000 psi requirement and 1400 psi pipeline MAOP, we would provide this pressure as follows. The pump station at the start of the pipeline will provide a discharge pressure of 1000 psi, which will be consumed by friction loss in the pipeline and at some point (roughly halfway) along the pipeline the pressure will drop to zero. At this location we boost the liquid pressure to 1000 psi using an intermediate booster pump station. We have assumed that the pipeline is essentially on a flat elevation profile.

This pressure of 1000 psi will be sufficient to take care of the friction loss in the second half of the pipeline length. The liquid pressure will reduce to zero at the end of the pipeline. Since the liquid pressure at any point along the pipeline must be above the vapor pressure of the liquid at the flowing temperature, and the intermediate pumps require certain minimum suction pressure, we cannot allow the pressure at any point to drop to zero. Accordingly, we will locate the second pump station at a point where the pressure has dropped to a suitable suction pressure, such as 50 psi. The minimum suction pressure required is also dictated by the particular pump and may have to be higher than 50 psi, to account for any restrictions and suction piping losses at the pump station. For the present, we will assume 50 psi suction pressure is adequate for each pump station. Hence, starting with a discharge pressure of 1050 psi (1000+50) we will locate the second pump station (intermediate booster pump) along the pipeline where the pressure has dropped to 50 psi. This pump station will then boost the liquid pressure back up to 1050 psi and will deliver the liquid to the pipeline terminus at 50 psi. Thus each pump station provides 1000 psi differential pressure (discharge pressure minus suction pressure) to the liquid, together matching the total pressure requirement of 2000 psi at 5000 gal/min flow rate.

Note that in the above analysis we ignored pipeline elevations and assumed that the pipeline profile is essentially flat. With elevations taken into account, the location of the intermediate booster pump will be different from that of a pipeline along a flat terrain.

Hydraulic balance is when each pump station supplies the same amount of energy to the liquid. Ideally pump stations will be located at hydraulic centers. This will result in the same horsepower (HP) being added to the liquid at each pump station. For a single flow rate at the inlet of the pipeline (no intermediate injections or deliveries), the hydraulic centers will also result in the same discharge pressures at each pump

station. Due to topographic conditions it may not be possible to locate the intermediate pump station at the locations desired for hydraulic balance. For example, calculations may show that three pump stations are required to handle the flow rate and that the two intermediate pump stations are to be located at milepost 50 and milepost 85. The location of milepost 50, when investigated in the field, may be found to be in the middle of a swamp or a river. Hence we will have to relocate the pump station to a more suitable location after field investigation. If the revised location of the second pump station were at milepost 52, then obviously hydraulic balance would no longer be valid. Recalculations of the hydraulics with the newly selected pump station locations will show hydraulic imbalance and all pump stations will not be operating at the same discharge pressure or providing the same amount of HP to the liquid at each pump station. However, while it is desirable to have all pump stations balanced, it may not be practical. Balanced pump station locations afford the advantage of using identical pumps and motors and the convenience of maintaining a common set of spare parts (pump rotating elements, mechanical seal, etc.) at a central operating district location.

In [Chapter 5](#) we discussed how the location of an intermediate pump station can be calculated from given data on pump station suction pressure, discharge pressure, etc. In Section 5.2 we presented a formula to calculate the discharge pressure for a pipeline system with two pump stations given the total pressure required for a particular flow rate. We will expand on that discussion by presenting a method to calculate the pump station pressures for hydraulic balance.

Figure 6.1 shows a pipeline with varying elevation profile but no significant controlling peaks along the pipeline. The total pressure P_T was

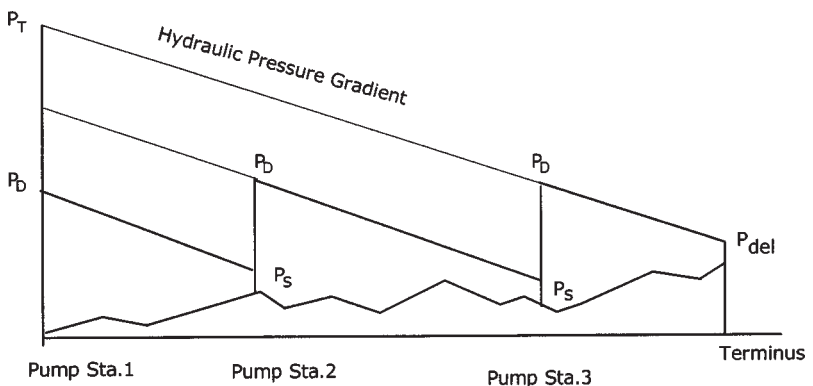


Figure 6.1 Hydraulic gradient: multiple pump stations.

calculated for the given flow rate and liquid properties. The hydraulic gradient with one pump station at the total pressure P_T is as shown. Since P_T may be higher than the pipe MAOP, we will assume that three pump stations are required to provide the pressures needed within MAOP limits. Each pump station will discharge at pressure P_D . If P_S represents the pump station suction pressure and P_{del} the delivery pressure at the pipeline terminus, we can state the following, using geometry:

$$P_D + (P_D - P_S) + (P_D - P_S) = P_T \quad (6.1)$$

Since the above is based on one origin pump station and two intermediate pump stations, we can extend the above equation for N pump stations as follows:

$$P_D + (N-1) \times (P_D - P_S) = P_T \quad (6.2)$$

Solving for N we get

$$N = (P_T - P_S) / (P_D - P_S) \quad (6.3)$$

Equation (6.3) is used to estimate the number of pump stations required for hydraulic balance given the discharge pressure limit P_D at each pump station. Solving Equation (6.3) for the common pump station discharge pressure,

$$P_D = (P_T - P_S) / N + P_S \quad (6.4)$$

As an example, if the total pressure calculated is 2000 psi and the suction pressure is 25 psi, the number of pump stations required with 1050 psi discharge pressure is

$$N = (2000 - 50) / (1050 - 50) = 1.95$$

Rounding up to the nearest whole number, we can conclude that two pump stations are required. Using Equation (6.4), the discharge pressure P_D at which each pump station will operate is:

$$P_D = (2000 - 50) / 2 + 50 = 1025 \text{ psi}$$

Once we have calculated the discharge pressure required for hydraulic balance, as above, a graphical method can be used to locate the pump stations along the pipeline profile. First the pipeline profile (milepost versus elevation) is plotted and the hydraulic gradient superimposed upon it by drawing the sloped line starting at P_T at A and ending at P_{del} at D, as shown in [Figure 6.1](#). Note that the pressure must be converted to feet of head, since the pipe elevation profile is plotted in feet. Next, starting at the first pump station A at discharge pressure P_D , a line is drawn parallel to the hydraulic gradient. The location B of the second pump station will be

established at a point where the hydraulic gradient between A and B meets the vertical line at the suction pressure P_s . The process is continued to determine the location C of the third pump station.

In the above analysis we have made several simplifying assumptions. We assumed that the pressure drop per mile was constant throughout the pipeline, meaning the pipe internal diameter was uniform throughout. With variable pipe diameter or wall thickness, the hydraulic gradient slopes between pump station segments may not be the same.

6.2 Telescoping Pipe Wall Thickness

On examining the typical hydraulic gradient shown in Figure 6.2, it is evident that under steady-state operating conditions the pipe pressure decreases from pump station to the terminus in the direction of flow. Thus, the pipeline segment immediately downstream of a pump station will be subject to higher pressures such as 1000 to 1200 psi while the tail end of that segment before the next pump station (or terminus) will be subject to lower pressures in the range of 50 to 300 psi. If we use the same wall thickness throughout the pipeline, we will be underutilizing the downstream portion of the piping. Therefore, a more efficient approach would be to reduce the pipe wall thickness as we move away from a pump station toward the suction side of the next pump station or the delivery terminus.

The higher pipe wall thickness immediately adjacent to the pump station will be able to withstand the higher discharge pressure and, as the pressure reduces down the line, the lower wall thickness would be designed to withstand the lower pressures as we approach the next pump station or delivery terminus. This process of varying the wall thickness to compensate

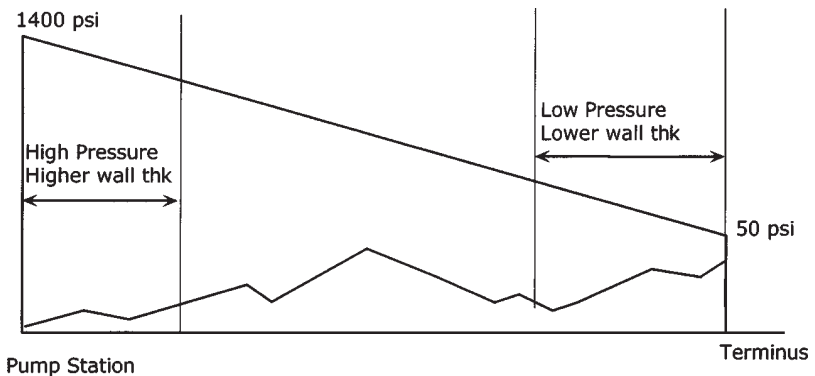


Figure 6.2 Telescoping pipe wall thickness.

for reduced pipeline pressures is referred to as telescoping pipe wall thickness.

A note of caution regarding wall thickness tapering would be appropriate here. If a pipeline has two pump stations and the second pump station is shut down for some reason, the hydraulic gradient is as shown in Figure 6.3. It can be seen that portions of the pipeline on the upstream side of the second pump station will be subject to higher pressure than when the second pump station was online. Therefore, wall thickness reductions (telescoping) implemented upstream of a pump station must be able to handle the higher pressures that result from shut-down of an intermediate pump station.

6.3 Change of Pipe Grade: Grade Tapering

In the same way that pipe wall thickness can be varied to compensate for lower pressures as we approach the next pump station or delivery terminus, the pipe grade may also be varied. Thus the high-pressure sections may be constructed of X-52 grade steel whereas the lower-pressure section may be constructed of X-42 grade pipe material, thereby reducing the total cost. This process of varying the pipe grade is referred to as grade tapering. Sometimes a combination of telescoping and grade tapering is used to minimize pipe cost. It must be noted that such wall thickness variation and pipe grade reduction to match the requirements of steady-state pressures may not always work. Consideration must be given to increased pipeline pressures when intermediate pump stations shut-down and or under upset conditions such as pump start up, valve closure, etc. These transient conditions cause surge pressures in a pipeline and therefore must be taken into account when selecting optimum wall thickness and pipe grade. This is illustrated in the modified hydraulic pressure gradient under transient conditions as depicted in Figure 6.4.

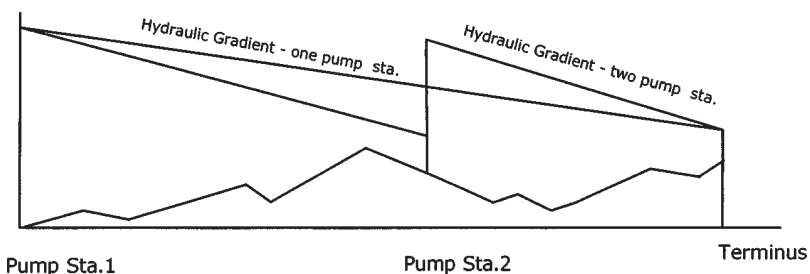


Figure 6.3 Hydraulic gradient: pump station shut-down.

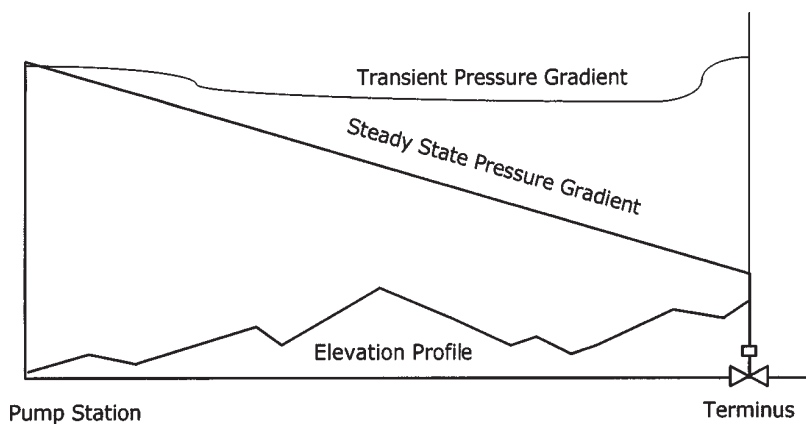


Figure 6.4 Hydraulic gradient: steady-state versus transient.

6.4 Slack Line and Open Channel Flow

Generally most pipelines flow full with no vapor space or a free liquid surface. However, under certain topographic conditions with drastic elevation changes, we may encounter pipeline sections that are partially full, called open channel flow or slack line conditions. Slack line operation may be unavoidable in some water lines, refined product and crude oil pipelines. Such a flow condition cannot be tolerated with high vapor pressure liquids and in batched pipelines. In the latter there would be intermingling of batches with disastrous consequences.

Consider a long pipeline with a very high peak at some point between the origin A and the terminus B as shown in [Figure 6.5](#). Due to the high elevation point at C, the pressure at A must be sufficient to take care of the friction loss between A and C, and the pressure head due to elevation difference between A and C, and the minimum pressure required at the top of the hill at C to prevent vaporization of liquid. Once the liquid reaches the peak at C with the required minimum pressure, the elevation difference between C and B helps the liquid gain pressure as it flows down the hill from C to the terminus at B. The frictional pressure drop between C and B has an opposite effect to the elevation and hence the resultant pressure at the terminus B will be the difference between the elevation head and the friction head. If the elevation head between C and B is sufficiently high compared with the frictional pressure drop between C and B, the final delivery pressure at B will be higher than the minimum required at the terminus. If the delivery at B is into an atmospheric storage tank, the hydraulic gradient will be modified as shown in the dashed line for slack line conditions.

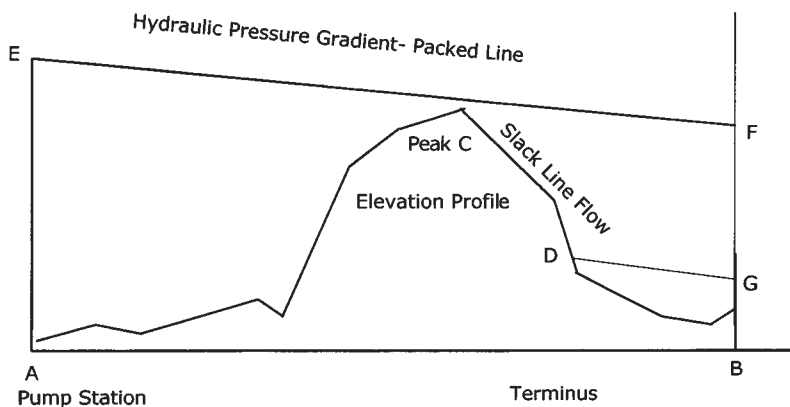


Figure 6.5 Hydraulic gradient: slack line versus packed line.

The upper hydraulic gradient depicts a packed line condition where the delivery pressure at B represented by point F is substantially higher than that required for delivery into a tank represented by point G. The lower hydraulic gradient shows that a portion of the pipeline between the peak at C and a point D will run in a partially full or slack line condition. Every point in the pipeline between C and D will be at zero gauge pressure. From D to B the pipe will run full without any slack. The slack line portion CD of the pipeline where the pipe is only partially full of the liquid is also referred to as open channel flow.

In this portion of the pipeline both liquid and vapor exists, which is an undesirable condition especially when pumping high vapor pressure liquids. Since a minimum pressure has to be maintained at the peak C, to prevent vaporization, the subsequent open channel flow in section CD of the pipeline defeats the purpose of maintaining a minimum pressure in the pipeline. In such instances, the pipeline must be operated in a packed condition (no slack line or open channel flow) by providing the necessary back pressure at B using a control valve, thus bringing the hydraulic gradient back to EF. The control valve at B should have an upstream pressure equal to the pressure that will produce the packed line hydraulic gradient showed in Figure 6.5. In crude oil and refined product pipelines where a single product is transported, slack line can be tolerated. However, if the pipeline is operated in batched mode with multiple products flowing simultaneously, slack line cannot be allowed since intermingling and consequently degradation of the different batches would occur. A batched pipeline must therefore be operated as a tight line by using a control valve to create the necessary back pressure to pack the line.

In [Figure 6.5](#), the back pressure valve would maintain the upstream pressure corresponding to point F on the hydraulic gradient. Downstream of the valve the pressure would be lower for delivery into a storage tank.

6.5 Batching Different Liquids

Batching is the process of transporting multiple products simultaneously through a pipeline with minimal mixing. Some commingling of the batches is unavoidable at the boundary or interface between contiguous batches.

For example, gasoline, diesel, and kerosene may be shipped through a pipeline in a batched mode from a refinery to a storage terminal. Batched pipelines have to be run in turbulent mode, with velocities sufficiently high to ensure a Reynolds number over 4000. If flow were laminar ($R < 2100$), the product batches would intermingle, thereby contaminating or degrading the products. Also, as discussed earlier, batched pipelines must be run in packed conditions (no slack line or open channel flow) to avoid contamination or intermingling of batches in pipelines with drastic elevation changes such as the one shown in [Figure 6.5](#).

In a batched pipeline, the total frictional pressure drop for a given flow rate will be calculated by adding the individual pressure drops for each product, considering its specific gravity, viscosity, and the batch length. We will illustrate this using an example.

Example Problem and Solution 6.1

Consider a 16 in. pipeline, 0.250 in. wall thickness, 100 miles long from Douglas Refinery to Hampton Terminal used to ship three products at a flow rate of 4000 bbl/hr as shown in [Figure 6.6](#). The three batches shown represent an instantaneous snapshot condition.

First we calculate the total liquid volume in the pipeline. This is referred to as the line fill volume. The line fill volume can be calculated using the following equation:

$$\text{Line fill volume} = 5.129L(D)^2 \quad (6.5)$$

where

D=Pipe internal diameter, in.

L=Pipe length, miles

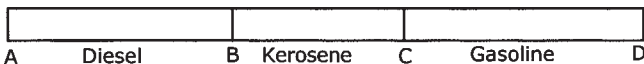


Figure 6.6 Batched pipeline.

In our example,

$$\text{Line fill} = 5.129(100)(15.5)^2 = 123,224 \text{ bbl}$$

Assuming the following physical properties and batch sizes for the three liquids, we can calculate the pressure drop for each liquid batch at the given flow rate:

Product	Specific gravity	Viscosity (cSt)	Batch size (bbl)
Diesel	0.85	5.9	50,000
Kerosene	0.82	2.7	30,000
Gasoline	0.74	0.7	43,224

Using the Colebrook-White equation the frictional pressure drops in the different batch segments are calculated to be as follows:

Diesel:	10.17 psi/mile	Batch length: 40.58 miles
Kerosene:	8.58 psi/mile	Batch length: 24.35 miles
Gasoline:	6.56 psi/mile	Batch length: 35.07 miles

In the above, for each liquid the pipeline length that represents the batch volume is shown. Thus the diesel batch will start at milepost 0.0 and end at milepost 40.58. Similarly the kerosene batch will start at milepost 40.58 and end at milepost $(40.58 + 24.35) = 64.93$. Finally, the gasoline batch will start at milepost 64.93 and end at milepost $(64.93 + 35.07) = 100.0$ for the snapshot configuration shown in [Figure 6.6](#).

The batch lengths calculated above are based on 1232.24 bbl per mile of 16 in. pipe calculated using Equation 6.5. The total frictional pressure drop for the entire 100 mile pipeline is obtained by adding up the individual frictional pressure drops for each product as follows:

$$\begin{aligned} \text{Total pressure drop} &= 10.17(40.58) + 8.58(24.35) \\ &\quad + 6.56(35.07) = 851.68 \text{ psi.} \end{aligned}$$

In addition to the frictional pressure drop, the elevation head and the delivery pressure are combined to calculate the total pressure required at Douglas Refinery.

In a batched pipeline, there will be some intermixing of the two adjacent products, resulting in an interface of commingled product. The volume of this interface varies with the pipe size, Reynolds number, and pipe length. Several correlations have been developed to estimate the amount of contamination in products at the interface.

As the batches arrive at the destination, density measuring instruments or densitometers (or gravitometers) monitor the density (or gravity) and flow is switched from one tank to another as appropriate. The contaminated volumes at the batch interface are diverted into a slop tank and later blended into a lower-grade product.

When batching different products such as gasoline and diesel, flow rates vary as the batches move through the pipeline, due to the changing composition of liquid in the pipeline. We will discuss this further in [Chapter 7](#), when centrifugal pump performance is combined with system head curves.

In order to operate the batched pipeline economically, by minimizing pumping cost, there exists an optimum batch size for the various products in the pipeline system. An analysis needs to be made over a finite period, such as a week or a month, to determine the flow rates and pumping costs considering various batch sizes. The combination of batch sizes that results in the least total pumping cost, consistent with shipper and market demands, will then be the optimum batch sizes for that particular pipeline system.

6.6 Summary

We have covered hydraulic balance in pipelines with multiple pump stations and learned that hydraulic balance may not always be possible due to topographic conditions. We demonstrated an approach to determine the number of pump stations from the total pressure required, minimum pump suction pressure, and allowable pump discharge pressure based on pipeline MAOP. The advantages and cost implications of telescoping wall thickness and pipe grade tapering were discussed. Slack line and open channel flow may occur in certain cases but should be avoided in batched pipelines and when pumping high vapor pressure liquids. Also covered was the method of calculating hydraulics in a batched pipeline by analyzing a snapshot configuration of multiple products in a pipeline.

6.7 Problems

- 6.7.1 A pipeline 150 miles long from Beaumont pump station to a tank farm at Glendale is used to transport Alaskan North Slope crude oil (ANS crude). The pipe is 20 in. in outside diameter and constructed of X-52 steel. It is desired to operate the system at ANSI 600 pressure level (1440 psi). The pipeline profile is such that there are two peaks located between Beaumont and Glendale. The first peak occurs at milepost 65.0 at an elevation of 1500 ft. The second peak is located at

milepost 110.0 at an elevation of 2500 ft. Beaumont has an elevation of 350 ft and Glendale is situated at an elevation of 650 ft. During the initial phase of operation, 6000 bbl/hr of ANS crude will be pumped at a temperature of 60°F and delivered to Glendale tankage at a pressure of 30 psi. The specific gravity and viscosity of ANS crude at 60°F may be assumed to be 0.895 and 43 cSt respectively.

- (a) Determine the minimum wall thickness required to operate the pipeline system at a pressure of 1400 psi.
- (b) At a flow rate of 6000 bbl/hr, how many pump stations would be required?
- (c) During the second phase it is planned to expand the capacity of pipeline to 9000 bbl/hr. How many additional pump stations would be required?
- (d) Assuming 80% pump efficiency, calculate the total pumping HP required during the initial phase and under the expansion scenario. Use a minimum suction pressure of 50 psi at each pump station.

6.7.2 The pipeline described in Problem 6.7.1 is used to batch ANS crude along with a light crude (0.85 specific gravity and 15 cSt viscosity at 60°F). Determine the optimum batch sizes to reduce pumping costs based on a 30 day operation. Consider a flow rate of 6000 bbl/hr.

6.7.3 A batched pipeline, 60 km long, is used to ship three grades of refined products. An instantaneous configuration shows an 8000 m³ batch of gasoline followed by 10,000 m³ of diesel, the remainder of the line consisting of a batch of kerosene. The pipe size is 500 mm diameter and 10 mm wall thickness and it traverses essentially flat terrain. Assume 8 MPa for the operating pressure and a pipeline delivery pressure of 300 kPa. Calculate the total pressure drop in the pipeline at a flow rate of 1800 m³/hr. The specific gravity and viscosities of the three products at the flowing temperature of 20°C are:

	Specific gravity	Viscosity (cSt)
Gasoline	0.74	0.65
Kerosene	0.82	1.5
Diesel	0.85	5.9